# AGBO FRANKLIN EMEKA


## Full-Stack Software Engineer

Known professionally as **Frank Joe**

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## PROFESSIONAL SUMMARY


Full-Stack Software Engineer focused on building production-ready web applications and business systems. I take a project from requirements to a deployed product and care that it keeps working once real people depend on it. Experience across the whole stack with Next.js, TypeScript, Node.js, PostgreSQL, Supabase and MongoDB, including authentication and role-based access, payment integrations with webhook verification, admin dashboards, REST APIs and real-time workflows. Delivered a hotel management system that is running in production for a real hotel.


## TECHNICAL SKILLS

**Frontend:** React, Next.js 14/15, TypeScript, JavaScript, Tailwind CSS, PWA (Service Workers, Web Push)

**Backend:** Node.js, Express, REST APIs, Authentication (JWT), Webhooks, Middleware, Rate Limiting

**Databases:** PostgreSQL, Supabase, MongoDB, Mongoose, MongoDB Atlas

**Payments:** Paystack integration, server-side payment verification, webhook confirmation

**Real-time:** LiveKit (video consultations), Pusher (WebSockets / push notifications)

**Tools & Infrastructure:** Vercel, Git, GitHub, Docker, AWS, Postman

**Security:** Role-based access control, protected route middleware, input validation, environment secrets

## WHAT I BUILD

SaaS platforms · Business management systems · Payment systems · Admin & analytics dashboards · AI-powered applications · APIs & backend systems

## WORK EXPERIENCE

### Full-Stack Software Engineer | Independent / Freelance | *2025 — Present*

*Nigeria · Remote*

- Design, build and deploy web applications and business systems directly for clients, working from business requirements to production.
- Delivered the **5G Hotel Management System**, now used by a real hotel: online booking, live room availability, Paystack payments confirmed by webhook, and receptionist, manager and CEO dashboards.
- Implemented authentication, role-based access and secure REST APIs, and designed database architectures for multi-role systems.
- Built administrative dashboards and real-time workflows, and deployed every application to production.
- Building **SmartCollege**, a school management platform with role-based portals, an AI assistant, AI-assisted results marking and automatic results generation.

### Developer | Erolite — Crypto Token Platform | *Contract*

- Contributed to core development of a crypto-based token system, building liquidity logic and transaction processing workflows.
- Implemented secure backend services for token operations and user account management.
- Delivered front-end and back-end features using Next.js and Node.js, collaborating on system architecture decisions.

### Frontend Editor & Management | CribMB — Medical Video & Chat Platform | *Employment*

- Managed and edited the frontend of a healthcare platform that lets patients consult doctors via live video and real-time chat.
- Maintained UI consistency, handled content updates and ensured a smooth experience across patient and doctor dashboards.
- Coordinated frontend changes with the development team, reviewing and managing interface updates aligned with product goals.

**Website Developer & Manager** | **Starlinx — Event Planning Company** | ***Contract***

- Designed, built and managed the full company website for a professional event planning business.
- Developed event showcase pages, booking and inquiry flows, and admin content management tools.
- Maintained site performance, branding consistency and regular content updates.

## PROJECTS

**5G Hotel Management System** — **Production system** | **fjoe.vercel.app/projects/5g-hotel**

Full-stack hotel booking and operations platform. Specific and multi-room booking with temporary room holds, Paystack payment with server-verified webhook confirmation, unique single-use booking codes, email confirmation, automated stale-booking cleanup, staff clock-in/out and working-hour tracking, and role-based receptionist, manager and CEO dashboards.

*Stack: Next.js, TypeScript, Supabase, PostgreSQL, Paystack*

**SmartCollege** — **School management platform**

Role-based school portals with an AI assistant, AI-assisted results marking and automatic results generation.

**PharmaLab** — **Pharmacy & lab management**

Multi-branch pharmacy inventory and lab management system.

**MedCore HMS** — **Hospital Management System** | **medcore-delta.vercel.app**

Hospital management system with admin dashboard, patient portal, blog, doctor video consultations, PWA support and real-time push notifications.

*Stack: Next.js 15, TypeScript, Tailwind CSS, MongoDB Atlas, LiveKit, Pusher, Vercel*

**FabSmile** — **Dental Clinic Management App** | **fabsmile.vercel.app**

Dental clinic platform with appointment booking, patient records, admin controls and secure role-based JWT authentication with server-side access validation.

*Stack: Next.js 14, TypeScript, Tailwind CSS, MongoDB Atlas, JWT, Vercel*

**MedSpa** — **Dental Clinic Management Web App**

Web application for managing a dental clinic.

**Lezadental** — **Dental Clinic Management Web App**

Web application for managing a dental clinic.

**WHFX** — **Investment Web App**

Investment web application.

**StockPilot** — **SaaS platform**

Multi-user SaaS application.

**FancyDoc** — **Document Management Tool** | **fancy-doc.vercel.app**

Web application for creating and managing documents with a clean UI, real-time updates and full CRUD functionality.

*Stack: Next.js, TypeScript, Tailwind CSS, MongoDB, Vercel*

**Eve Circle** — **Ovulation & Fertility Tracker** | **eve-circle.vercel.app**

Women's health web app that calculates ovulation windows and fertile days with a calendar-based UI.

*Stack: Next.js, TypeScript, Tailwind CSS, Vercel*

## CERTIFICATIONS & TRAINING

**Next.js & Full Stack Development** | **NextCollege (2025)**

- Mastered the Next.js App Router, Server Components and full CRUD application development.
- Built and deployed production Next.js applications with authentication and database integration.

## EDUCATION

**B.Sc. Medical Rehabilitation** | **University of Nigeria, Nsukka**